(34769)
35088
34447
4721
4721

VERTICAL | FACE

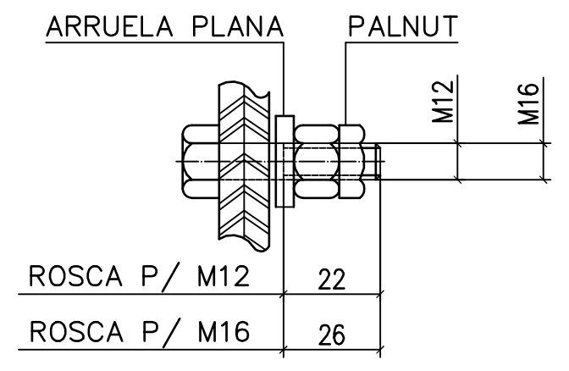
tg a = 0.1370971 | tg g = 0.1358265
2tg a = 0.2741941 | sec g = 1.0091823
sec a = 1.0093541
sec b = 1.0186222

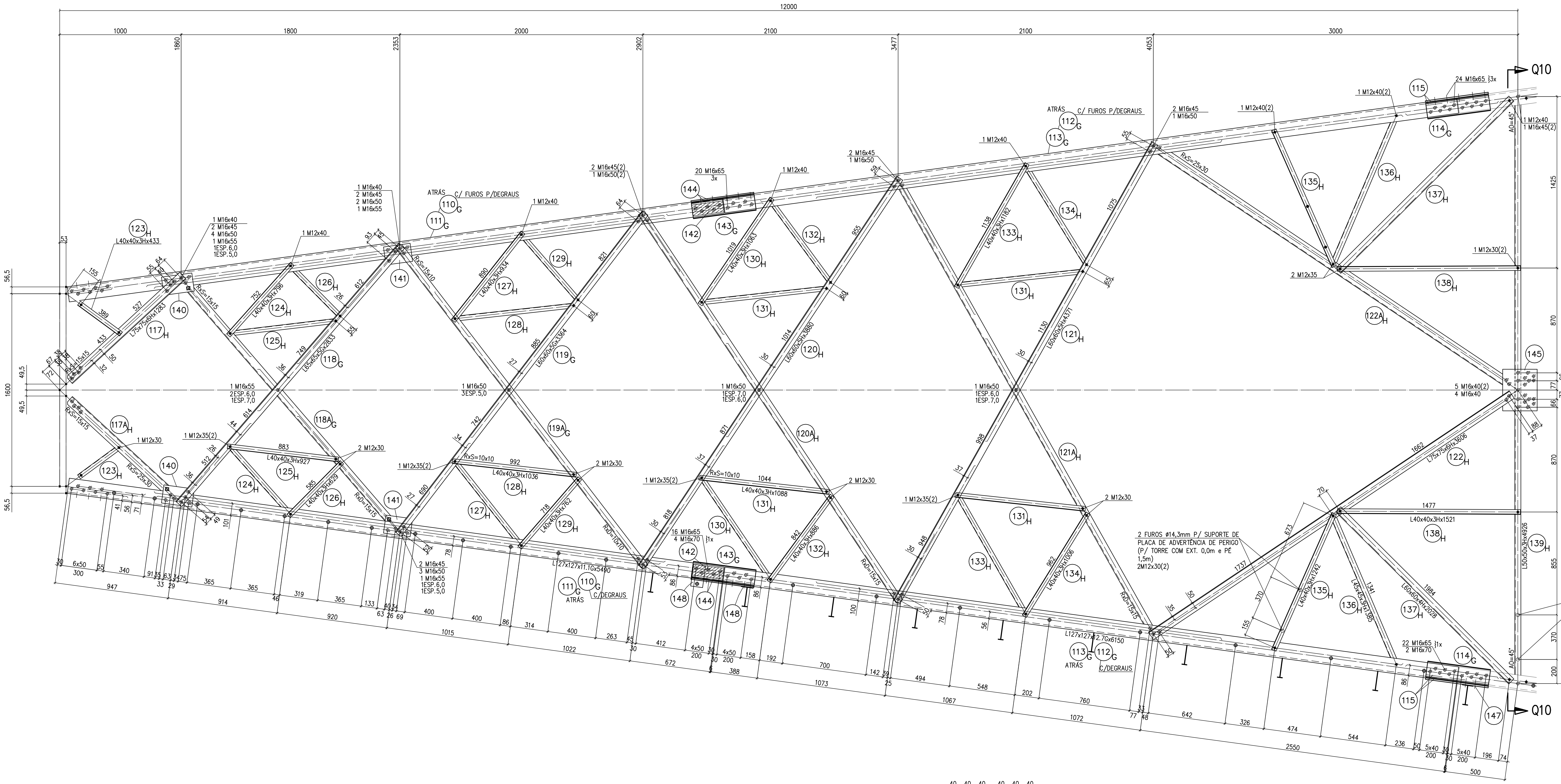
(34769)
35088
34447
4721
4721

VERTICAL | FACE

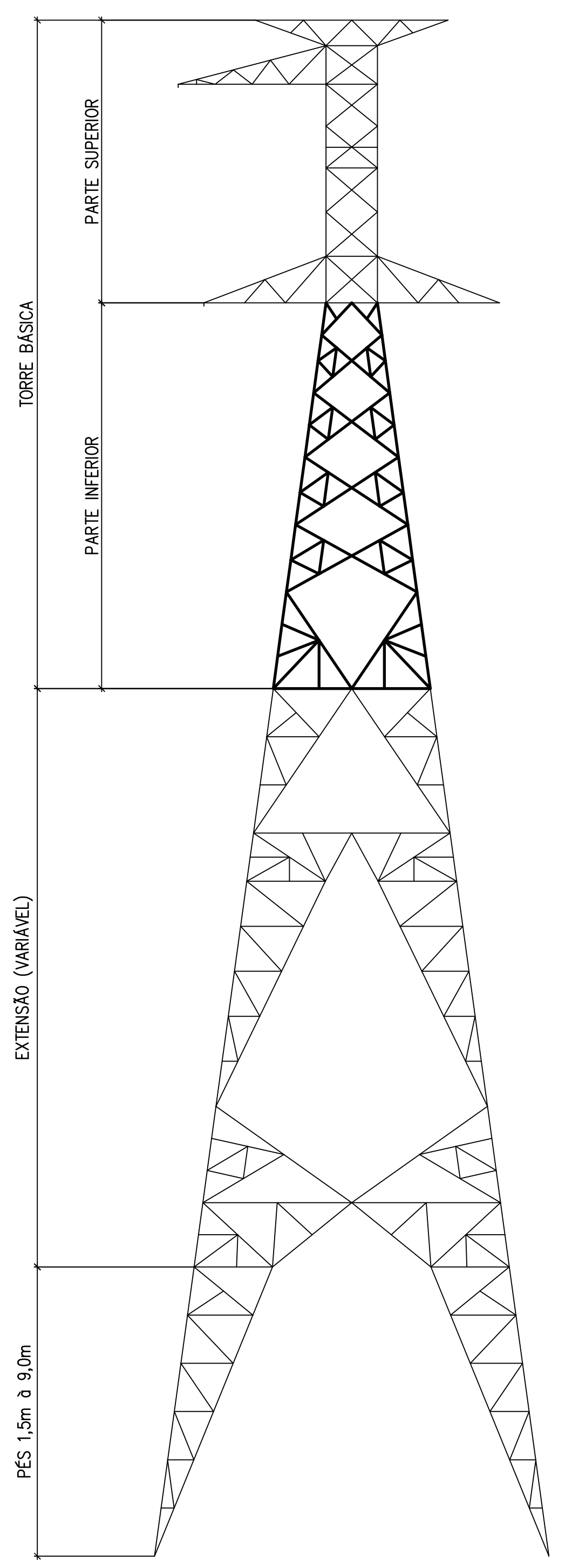
tg a = 0.1370971 | tg g = 0.1358265
2tg a = 0.2741941 | sec g = 1.0091823
sec a = 1.0093541
sec b = 1.0186222

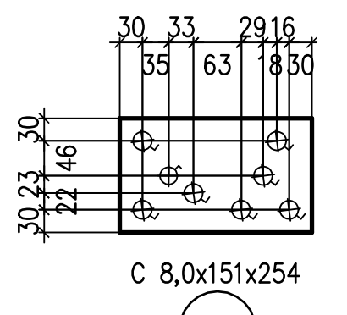
PARAFUSOS e ACESSÓRIOS	QUANTIDADES PARA 1 (UMA) TORRE	
PARAFUSOS / COMPRIMENTOS	M12	M16
30	42	–
35	24	–
40	28	24
45	–	44
50	–	44
55	–	10
60	–	–
65	–	170
70	–	6
75	–	–
ARRUELA PLANA ESP. 3,0	124	324
"PALNUT"	94	298
ESPESSURAS		
# 3,0	–	–
# 4,0	–	–
# 5,0	–	12
# 6,0	–	14
# 7,0	–	6
# 8,0	–	–
PARAFUSO DEGRAU PD–18M	21	

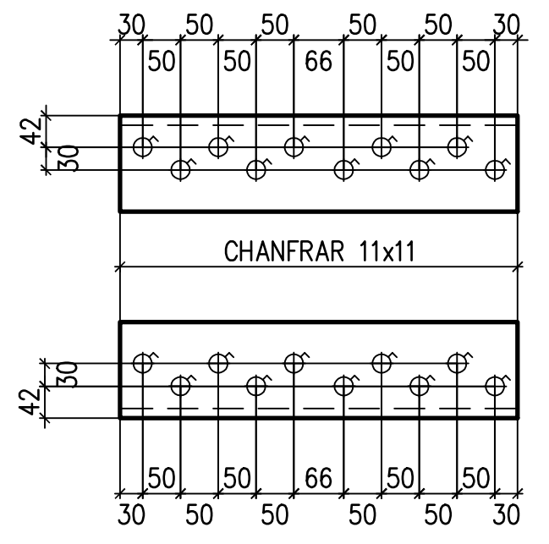


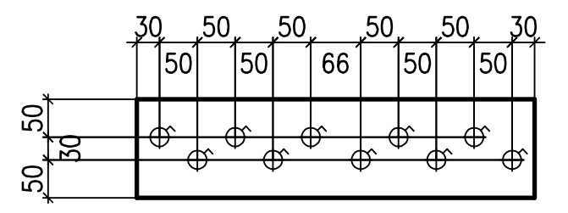

12000
1000
1800
2000
2100
2100
3000
Q10
Q10
ATRÁS C/ FUROS P/DEGRAUS
C/DEGRAUS
ATRÁS
L40x40x3Hx433
1 M16x40
2 M16x45
4 M16x50
1 M16x55
1ESP. 6,0
1ESP. 5,0
1 M16x55
2ESP. 6,0
1ESP. 7,0
1 M16x50
3ESP. 5,0
1 M16x50
1ESP. 7,0
1ESP. 6,0
5 M16x40(2)
4 M16x40
20 M16x65 3x
24 M16x65 3x
22 M16x65 1x
2 M16x70
16 M16x65
4 M16x70 1x
L127x127x11,7Gx5490
L127x127x12,7Gx6150
L40x40x3Hx1521
L50x50x3Hx1926
2 FUROS ø14,3mm P/ SUPORTE DE PLACA DE ADVERTÊNCIA DE PERIGO (P/ TORRE COM EXT. 0,0m e PÉ 1,5m) 2M12x30(2)
2 FUROS ø14,3mm P/ SUPORTE DE PLACA DE ADVERTÊNCIA DE PERIGO (P/ TORRE COM EXT. 0,0m e PÉ 3,0m) 2 M12x30(2)
110 111 112 113 114 115 117 117A 118 118A 119 119A 120 120A 121 121A 122 122A 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 147 148

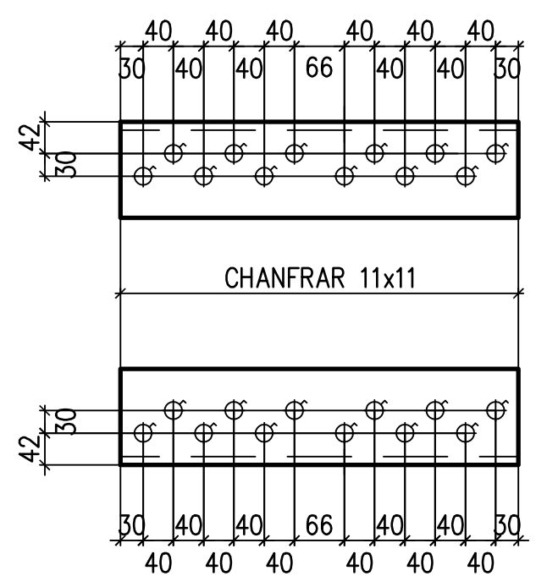
FACE TRANSVERSAL

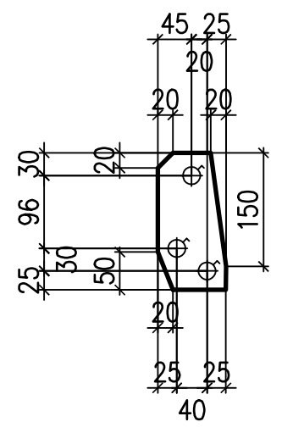


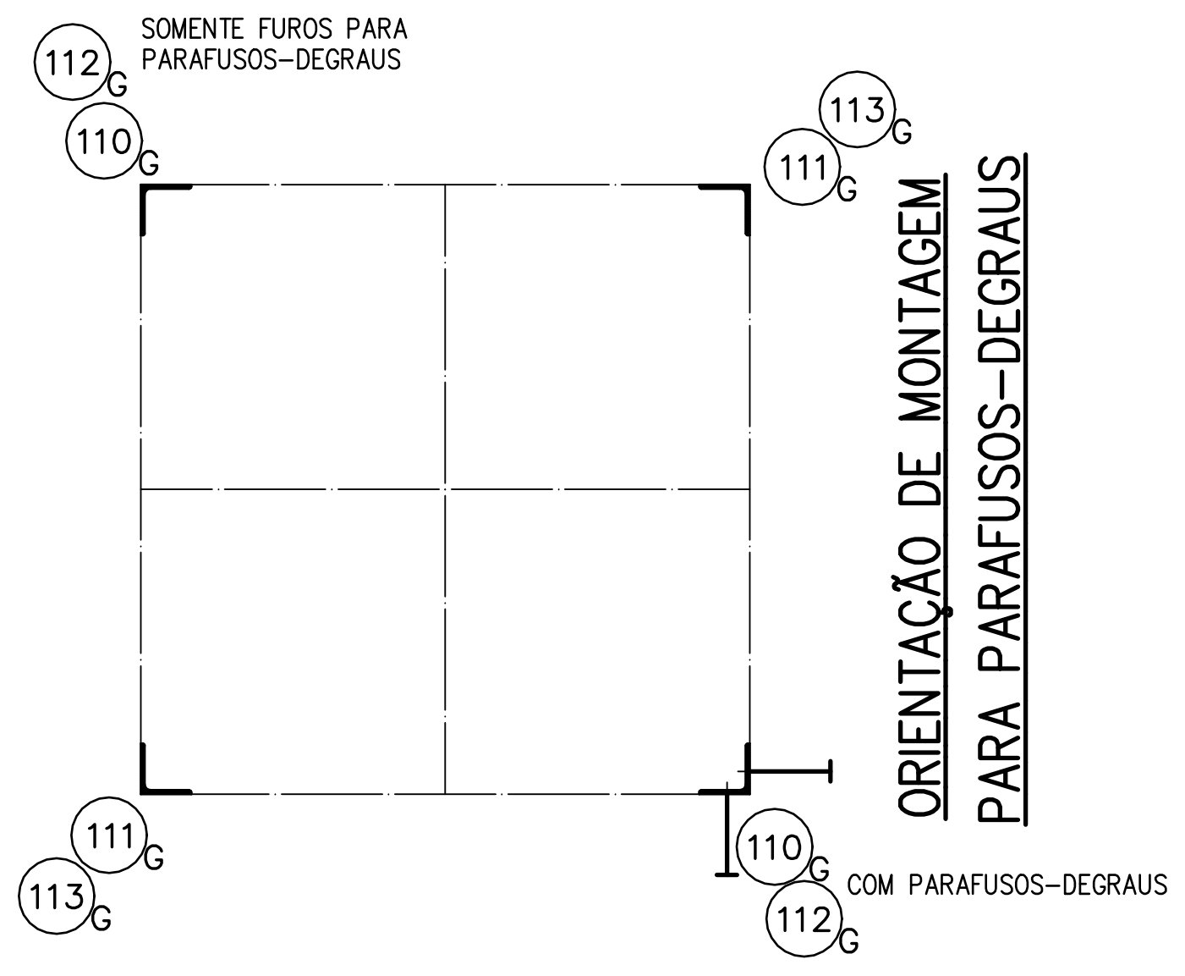

C 8,0x151x254
(140)


L 127x127x12,7Gx526
(143)G

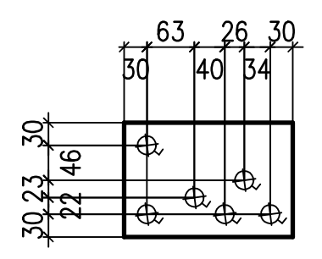

C 12,5x130x526
(144)

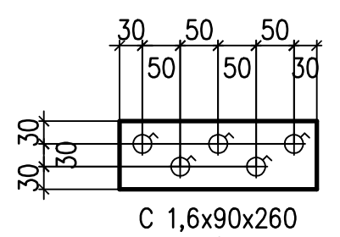

L 127x127x12,7Gx526
(114)G

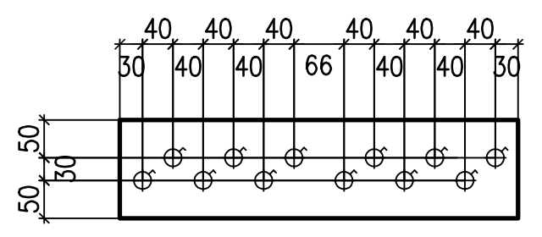

C 4,8x90x181
(147)

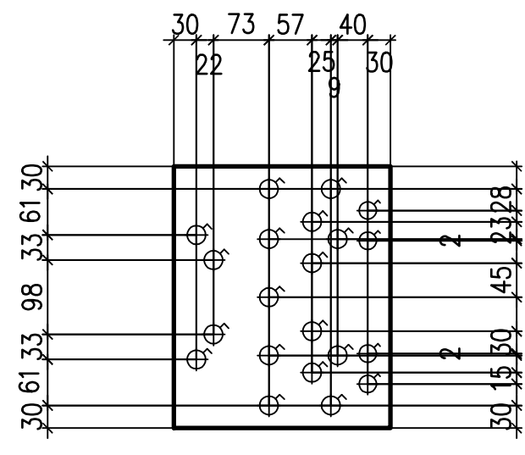

SOMENTE FUROS PARA PARAFUSOS-DEGRAUS
112G 110G 111G 113G
111G 113G
110G 112G COM PARAFUSOS-DEGRAUS

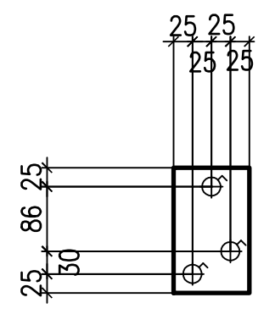
ORIENTAÇÃO DE MONTAGEM PARA PARAFUSOS-DEGRAUS


C 8,0x151x223
(141)


C 1,6x90x260
(142)


C 12,5x130x526
(115)


C 8,0x286x346
(145)


C 4,8x100x166
(148)

POS.: 110 : 115, 117 : 145, 147, 148, 117A, 118A, 119A, 120A, 121A, 122A

REV.	ALTERAÇÃO	DATA	DES.	VERIF.	APROV.
00	APROVADO	14/09/2026	M.D.R.	H.D.G.	H.D.G.
00B	CONFORME COMENTÁRIOS	28/08/2026	M.D.R.	H.D.G.	H.D.G.
00A	EMISSÃO INICIAL	20/03/2026	F.A.F.	L.C.C.	H.D.G.

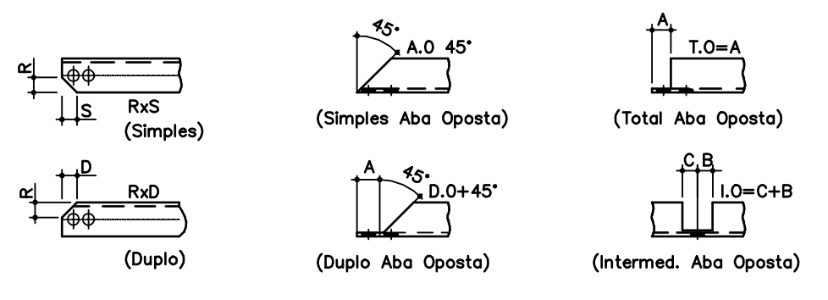
FINALIDADE
PARA APROVAÇÃO

SIMBOLOGIA DOS PARAFUSOS

- Parafusos M12 – Furos ø13,5 – A=22
- Parafusos M16 – Furos ø17,5 – A=25
- Parafusos M20 – Furos ø21,5 – A=30
- Parafusos M24 – Furos ø26,0 – A=40
- Parafusos Degraus PD16 – Furos ø17,5
- Furos Conforme ø indicado

A P/ Treliças

NOMENCLATURA DE RECORTES



TORQUE PARA APERTO DOS PARAFUSOS

ø	Min. (kg.m)	Max. (kg.m)
M12	3,5	4,5
M16	7,5	11,5
M20	13,0	22,0

REFERÊNCIAS

BNAT_ADR2451–D4002	ESPECIFICAÇÕES e NOTAS GERAIS
BNAT_ADR2451–D4003	TORRE BÁSICA – PARTE SUPERIOR: PARA–RAIOS E MÍSULA SUPERIOR
BNAT_ADR2451–D4004	TORRE BÁSICA: PARTE SUPERIOR
BNAT_ADR2451–D4005	TORRE BÁSICA – PARTE SUPERIOR: MÍSULA INFERIOR
BNAT_ADR2451–D4007	TORRE BÁSICA – PARTE INFERIOR: FACE LONGITUDINAL e CORTE Q10–Q10

HENRIQUE GONTIJO
RESPONSÁVEL TÉCNICO
CREA 108382/MG

LOTE 05 - ANEEL 01/2024
Nº DOCUMENTO: LTRH1.06-230-PE-ME.DMT-0057

AXIA ENERGIA
CONSAG CONSÓRCIO CONSTRUTOR SÃO FRANCISCO
AG ENGENHARIA
CR GONTIJO ENGENHARIA DE PROJETOS

Nº EMPREENDIMENTO: NOVA ERA INTEGRAÇÃO
UNIDADE: LOTE 5 - LT 230 kV
LOCAL: BOM NOME - BOM NOME II C1
CÓDIGO INSTALAÇÃO: -
TAG: -
REVISÃO: 00A
FOLHA: 01/01
Nº CONTRATO: -
Nº LEGADO: -

TÍTULO: TORRE DE ANCORAGEM MEIO DE LINHA PESADA e FIM DE LNHA TIPO BNAT
TORRE BÁSICA - PARTE INFERIOR: FACE TRANSVERSAL

Nº CR GONTIJO: BNAT_ADR2451-D4006
ESCALA: 1/10 - 1/15
REVISÃO: R- 00

Formato A0 (1189x841mm)